

Gaz parfait libre

Distribution des vitesses - Maxwell

$$\frac{1}{2} k_B T = \left\langle \frac{mv_x^2}{2} \right\rangle = \frac{m}{2} \langle v_x^2 \rangle = \frac{m}{2} \int_{-\infty}^{+\infty} v_x^2 \frac{1}{\sqrt{\pi}} \exp(-Bv_x^2) dv_x$$

$$\int_{-\infty}^{+\infty} x^2 \exp(-ax^2) dx = \frac{\sqrt{\pi}}{2 a^{3/2}}$$

$$\Leftrightarrow k_B T = m \sqrt{\frac{B}{\pi}} \cdot \frac{\sqrt{\pi}}{2 \cdot B^{3/2}} = \frac{m}{2B}$$

$$\Leftrightarrow B = \frac{m}{2k_B T}$$

$$p(v_x) = \sqrt{\frac{m}{2\pi k_B T}} \exp\left(-\frac{mv_x^2}{2k_B T}\right)$$

$$p(v_x) = A \exp(-B(v_x - 0)^2) \quad A=?; B=?$$

$$1 = \int_{-\infty}^{+\infty} p(v_x) dv_x = \int_{-\infty}^{+\infty} A \exp(-Bv_x^2) dv_x = A \frac{1}{\sqrt{B}} \int_{-\infty}^{+\infty} \exp(-y^2) dy = A \cdot \frac{\sqrt{\pi}}{\sqrt{B}}$$

$$\int_{-\infty}^{+\infty} \exp(-x^2) dx = \sqrt{\pi}$$

$$y^2 = Bv_x^2 \Leftrightarrow y = \sqrt{B} v_x$$

$$dy = \sqrt{B} dv_x \Leftrightarrow dv_x = \frac{1}{\sqrt{B}} dy$$

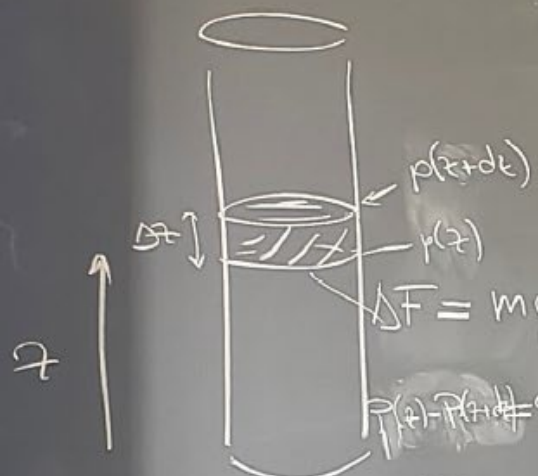
$$\Leftrightarrow A = \sqrt{\frac{B}{\pi}}$$

$$p(v_x) = \sqrt{\frac{m}{2\pi k_B T}} \exp\left(-\frac{mv_x^2}{2k_B T}\right)$$

Gaz parfait dans un champ de force
externe conservatif

Distribution des particules & loi barométrique

$p(z) = ?$ gaz parfait : $PV = N k_B T$
 isotherme : $P = \frac{V}{N} (k_B T) = Ck \cdot n$
 $dP = k_B T \cdot dn$



$\Delta F = mg = g \cdot m_u \cdot \Delta N = g \cdot m_u \cdot \Delta V \cdot n(z)$
 $p(z) - p(z+\Delta z) = g \cdot m_u \cdot \Delta z \cdot n(z) \cdot \frac{1}{\Delta z}$

$p(z+\Delta z) - p(z) = dP = -m_u \cdot g \cdot n(z) dz$
 $\Rightarrow k_B T \cdot dn(z) = -m_u g n(z) dz$
 $dn(z) = -\frac{m_u g}{k_B T} \cdot n(z) dz$

$p(z) dz = \frac{n(z) dz}{N}$

$N = \int_0^\infty n_0 \exp\left(-\frac{m_u g z}{k_B T}\right) dz = n_0 \left(-\frac{k_B T}{m_u g}\right) \left[\exp(-\infty) - \exp(0)\right] = n_0 \left(\frac{k_B T}{m_u g}\right)$

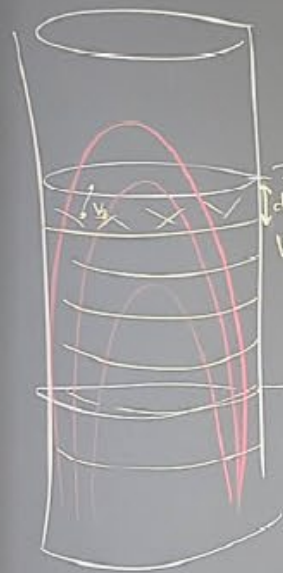
$\Rightarrow n_0 = \frac{N m_u g}{k_B T}$

$n(z) = n_0 \exp\left(-\frac{m_u g z}{k_B T}\right)$

$p(z) = P_0 \exp\left(-\frac{m_u g z}{k_B T}\right)$

$p(z) = \frac{n(z)}{N} = \frac{m_u g}{k_B T} \exp\left(-\frac{m_u g z}{k_B T}\right)$

Distribution des vitesses – Boltzmann (1/2)



$$z=h \quad (mgh = \frac{1}{2} mu^2)$$

$$V_x > u = \frac{1}{2} mu^2$$

$$u+2\Delta u$$

$$u+3\Delta u$$

$$u+4\Delta u$$

$$u+N\Delta u$$

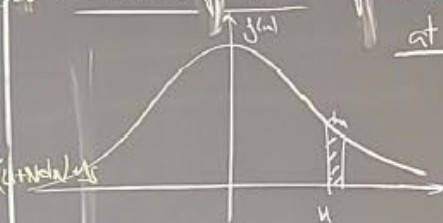
$$N \rightarrow \infty \quad \Delta u \rightarrow 0$$

$$\frac{n_{>u}(0)}{n_{>0}(0)} = \frac{n_{>0}(h)}{n_{>0}(0)} = \frac{\exp(-\frac{mgh}{k_B T})}{\exp(-\frac{m \cdot 0}{k_B T})} = \frac{\exp(-\frac{mu^2}{2k_B T})}{\exp(-\frac{m \cdot 0}{k_B T})} = \exp(-\frac{mu^2}{2k_B T})$$

number of molecules per unit area per second passing height h with a z -velocity $V_z > u$

$= \exp(-\frac{mu^2}{2k_B T}) \cdot$ total number of molecules passing at height $z=0$ with a z -velocity $V_z > 0$

We want $p(u)$ s.t. $p(u)du$ = fraction of molecules with speed $u \in [u, u+du]$ at all times



What is the number of molecules passing through a unit area per second with a velocity greater than u ?

$$\int_u^\infty p(v_z) dv_z \quad \text{passing area per second} \quad \int_u^\infty v_z p(v_z) dv_z = \exp(-\frac{mu^2}{2k_B T}) \cdot C$$

$$\lim_{N \rightarrow \infty, \Delta u \rightarrow 0} \frac{1}{\Delta t} \sum_{v_z=u}^{u+N\Delta u} S \cdot \Delta t \cdot v_z \cdot p(v_z) \Delta v_z = \int_u^\infty v_z p(v_z) dv_z$$

number of molecules in volume $S(\Delta t \cdot v_z)$ with speed $= v_z$

$$m g(h-0) = \frac{1}{2} m (V_z^2(h) - V_z^2(0)^2)$$

Distribution des vitesses – Boltzmann (2/2)

$$\frac{d}{du} \int_u^\infty v_z p(v_z) dv_z = \frac{d}{du} C \exp\left(-\frac{mu^2}{2k_B T}\right)$$

$$\Rightarrow \underbrace{0}_{\infty p(\infty)} + p(u) = C \exp\left(-\frac{mu^2}{2k_B T}\right) \left(+ \frac{d}{du} \frac{mu^2}{2k_B T}\right)$$

$$1 = \int_{-\infty}^{+\infty} p(v_z) dv_z = C \frac{m}{k_B T} \int_{-\infty}^{+\infty} \exp\left(-\frac{mv_z^2}{2k_B T}\right) dv_z = C \frac{m}{k_B T} \sqrt{\frac{2k_B T}{m}} \sqrt{\pi}$$

$$x^2 = \frac{m}{2k_B T} v_z^2$$

$$dv_z = \sqrt{\frac{2k_B T}{m}} dx$$

$$\int_{-\infty}^{+\infty} e^{-x^2} dx = \sqrt{\pi}$$

$$\Rightarrow 1 = C \cdot \frac{m}{k_B T} \sqrt{\frac{2k_B T}{m}} \sqrt{\pi} = \frac{\sqrt{2\pi m}}{\sqrt{k_B T}}$$

$$\Rightarrow C = \sqrt{\frac{k_B T}{2\pi m}}$$

$$p(v_z) = C \left(\frac{m}{k_B T}\right) \exp\left(-\frac{mv_z^2}{2k_B T}\right)$$

$$p(v_z) = \sqrt{\frac{k_B T}{2\pi m}} \cdot \frac{m}{k_B T} \exp(-\dots) = \sqrt{\frac{m}{2\pi k_B T}} \exp\left(-\frac{mv_z^2}{2k_B T}\right) \stackrel{!}{=} p(v_x) = p(v_y)$$

De la mécanique classique à la mécanique statistique

Equation de Newton – équation de Boltzmann

Newton: état donné par $\vec{x}_i$ et $\vec{v}_i$
 l'évolution temporelle est donnée par un système d'éq. différentielles :

p.ex. $m\ddot{x} + kx = 0 \Leftrightarrow \begin{cases} m\dot{v} + kx = 0 \\ \dot{x} = v \end{cases}$

Boltzmann: état donné par $P(\vec{x}, \vec{v}, t)$, p.ex. $P(\vec{x}, \vec{v}) \sim \exp\left(-\frac{m\vec{v}^2}{2k_B T}\right) \exp\left(-\frac{m\vec{x}^2}{2\sigma^2}\right) \cdot \frac{1}{\Delta x} \cdot \frac{1}{\Delta v} \cdot \exp\left(-\frac{mgz}{k_B T}\right)$

l'évolution temporelle est donnée par une éq. différentielle sur cette fonction de probabilité

$$\frac{\partial}{\partial t} P(\vec{x}, \vec{v}, t) + \vec{v} \cdot \vec{\nabla}_r P(\vec{x}, \vec{v}, t) + \frac{\vec{F}}{m} \cdot \vec{\nabla}_v P(\vec{x}, \vec{v}, t) = \left(\frac{\partial P}{\partial t} \right)_{\text{collisions}}$$

Fonction H de Boltzmann & Entropie

Fonction H de Boltzmann

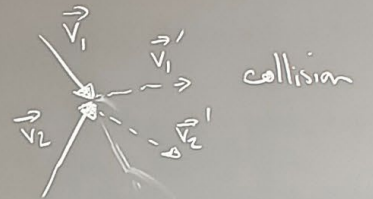
Equation de Boltzmann:
$$\left(\frac{\partial}{\partial t} + \vec{v} \cdot \vec{\nabla}_{\vec{r}} + \frac{\vec{F}}{m} \cdot \vec{\nabla}_{\vec{v}} \right) f(\vec{r}, \vec{v}, t) = \left(\frac{\partial f}{\partial t} \right)_{\text{collisions}} = N_a \int d^3v_2 \int d\Omega \sigma(\Omega) |\vec{v}_1 - \vec{v}_2| f(\vec{r}, \vec{v}_1, t) f(\vec{r}, \vec{v}_2, t) - f(\vec{r}, \vec{v}_1, t) f(\vec{r}, \vec{v}_2, t)$$

Cas particulier:

$\vec{F} = 0$ et $f(\vec{r}, \vec{v}, t) = \frac{1}{V} f(\vec{v}, t)$

$$\frac{\partial}{\partial t} f(\vec{v}, t) = \frac{N_a}{V} \int d^3v_2 \int d\Omega \sigma(\Omega) |\vec{v}_1 - \vec{v}_2| (f_1' f_2' - f_1 f_2)$$

$$\frac{\partial f(\vec{v}, t)}{\partial t} = 0 \Leftrightarrow f(\vec{v}_1') f(\vec{v}_2') = f(\vec{v}_1) f(\vec{v}_2) : \text{"état d'équilibre"}$$



NE PAS EFFACER!

Boltzmann définit la fonctionnelle H:
$$H(t) := \int d^3v n f(\vec{v}, t) \log[n f(\vec{v}, t)]$$

dont
$$\frac{dH}{dt} = n \int d^3v_1 \frac{\partial f_1}{\partial t} [1 + \log(n f_1(t))] = \dots = -\frac{n^2}{4} \int d^3v_1 \int d^3v_2 \int d\Omega \sigma(\Omega) |\vec{v}_1 - \vec{v}_2| (f_1 f_2 - f_1' f_2') [\log(f_1 f_2) - \log(f_1' f_2')] \leq 0$$

car $(x-y)(\log(x) - \log(y)) \geq 0$

$$\frac{dH}{dt} \leq 0 \text{ et } \frac{dH}{dt} \Leftrightarrow \frac{\partial f}{\partial t} = 0 \Leftrightarrow f(\vec{v}_1) f(\vec{v}_2) = f(\vec{v}_1') f(\vec{v}_2') \Leftrightarrow \text{on est à l'équilibre thermodynamique}$$